Sample Paper 2 — Answer Key

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **C) Potato** — Mango, Banana, Apple are fruits; Potato is a vegetable.
2. **B) School** — A doctor works in a hospital; a teacher works in a school.
3. **A) ECR** — The rule is "+2 letters": C → E, A → C, P → R.
4. **B) 15** — The pattern adds 3 each time: 3, 6, 9, 12, 15.
5. **C) Sana** — Order is Sana → Kabir → Mira, so Sana is at the front.
6. **A) North** — East rotated 90° anticlockwise is North.
7. **B) Saturday** — 15 days = 2 full weeks (14 days, same day: Sunday) + 1 more day back → Saturday.
8. **C) 15** — 11, 13, 17 are prime numbers; 15 = 3 × 5 is not.
9. **C) 80** — Each term doubles: 5, 10, 20, 40, 80.
10. **A) CJSE** — Each letter moves forward 1 step: B → C, I → J, R → S, D → E.

Section 2 — Mathematical Reasoning

11. **A) 10023** — 5678 + 4345 = 10023.
12. **A) 5825** — 9500 − 3675 = 5825.
13. **A) 3192** — 456 × 7 = 3192.
14. **A) 1008** — 8064 ÷ 8 = 1008.
15. **C) 8** — Factors of 16: 1,2,4,8,16. Factors of 24: 1,2,3,4,6,8,12,24. HCF = 8.
16. **C) 40** — Multiples of 5: 5,10,...,40. Multiples of 8: 8,16,...,40. First common multiple = 40.
17. **A) 3/5** — 3/5 = 0.6, which is greater than 1/2 = 0.5.
18. **B) 4000 g** — 1 kg = 1000 g, so 4 kg = 4000 g.
19. **B) 48 cm** — Perimeter = 2 × (15 + 9) = 48 cm.
20. **B) 2** — A rectangle (not a square) has 2 lines of symmetry.

Section 3 — Everyday Mathematics

21. **A) 235** — 850 − 375 − 240 = 235.
22. **B) 260 km** — 65 × 4 = 260 km.

23. **C) 23** — 276 ÷ 12 = 23 boxes exactly.
24. **B) 2 h 25 min** — 6:40 p.m. to 8:40 p.m. is 2 h; 8:40 p.m. to 9:05 p.m. is 25 more minutes.
25. **B) 6** — 9 L = 9000 mL; 9000 ÷ 1500 = 6 bottles.
26. **B) ₹240** — 4 × ₹60 = ₹240.
27. **B) 230 L** — 45 × 6 = 270 L used; 500 − 270 = 230 L left.
28. **B) 22 kg 200 g** — 12400 g + 9800 g = 22200 g = 22 kg 200 g.
29. **B) 6** — 18 ÷ 3 = 6 students per row.
30. **B) ₹35** — ₹185 − ₹150 = ₹35 profit.

Section 4 — Achievers Section (2 marks each)

31. **B) 131** — Number = (divisor × quotient) + remainder = (9 × 14) + 5 = 126 + 5 = 131.
32. **C) 260** — Larger number = 980 − 410 = 570; difference = 570 − 410 = 160; 160 + 100 = 260.
33. **B) ₹900** — Spent on book = 1/4 × 2000 = ₹500; remainder = ₹1500; spent on watch = 2/5 × 1500 = ₹600; left = 1500 − 600 = ₹900.
34. **C) 85** — Apples = 15; mangoes = 3 × 15 = 45; oranges = 15 + 10 = 25; total = 15 + 45 + 25 = 85.
35. **C) 21** — These are triangular numbers (1, 3, 6, 10, 15, 21, ...), where the 6th term = 6 × 7 ÷ 2 = 21.

Scoring guide (out of 40): 32+ is Level-2 cutoff range — strong. 24–31 is solid Level-1 performance — review the missed sections. Below 24 — revisit the relevant topics before attempting Sample Paper 3.